

## FINAL DIAGNOSIS:

### PART 1: THYROID, TOTAL THYROIDECTOMY (24 GRAMS) -

- PAPILLARY THYROID CARCINOMA, ENCAPSULATED FOLLICULAR VARIANT (3.6 CM) LOCATED IN THE LEFT LOBE, CONFINED TO THYROID; NO TUMOR CAPSULE INVASION.
- NO ANGIOLYMPHATIC INVASION OR EXTRATHYROIDAL EXTENSION.
- RESECTION MARGIN FREE OF TUMOR.
- ONE LEFT PERITHYROIDAL LYMPH NODE FREE OF TUMOR (0/1).
- PATHOLOGIC STAGE: pT2 N0.
- NODULAR THYROID HYPERPLASIA.

100-0-3  
carcinoma, papillary follicular  
variant 8340/3

### PART 2: LYMPH NODES, CENTRAL COMPARTMENT, DISSECTION -

- SEVEN LYMPH NODES FREE OF TUMOR (0/7).
- TISSUE ENTIRELY SUBMITTED FOR MICROSCOPIC EVALUATION.

Site: Thyroid, NOS C73.9

### CASE SYNOPSIS:

#### SYNOPTIC DATA - PRIMARY THYROID TUMORS

##### SPECIMEN TYPE:

Total Thyroidectomy

##### TUMOR SITE:

Left Lobe

##### TUMOR FOCALITY:

Unifocal

##### TUMOR SIZE (largest nodule):

Greatest Dimension: 3.6 cm

##### HISTOLOGIC TYPE:

Papillary carcinoma, encapsulated follicular variant

##### PRIMARY TUMOR (pT):

pT2

##### REGIONAL LYMPH NODES (pN):

pN0

Number of regional lymph nodes examined: 8

Number of regional lymph nodes involved: 0

##### DISTANT METASTASIS (pM):

Not applicable

##### EXTRATHYROIDAL EXTENSION:

Not identified

##### MARGINS:

Margins uninvolved by carcinoma

##### LYMPH-VASCULAR INVASION:

Not identified

##### ADDITIONAL PATHOLOGIC FINDINGS:

Other: nodular thyroid hyperplasia

hw  
10/7/12

### Addendum

### MOLECULAR ANATOMIC PATHOLOGY TESTING:

#### Block 1B:

- BRAF mutation IDENTIFIED (p.V600E, c.1799T>A).
- Mutations in NRAS61, HRAS61, KRAS12/13 and RET/PTC1 and RET/PTC3 rearrangements NOT identified.

#### NOTE:

Nucleic acids were extracted in the amount sufficient for testing.

#### BACKGROUND:

Mutations in either BRAF or RAS genes or RET/PTC rearrangements are found in more than 70% of papillary thyroid carcinomas (1). BRAF V600E (T1798A) mutation has been associated with more aggressive behavior of papillary carcinoma (2, 3). The association between BRAF V600E mutation and features of tumor aggressiveness have also been observed in papillary microcarcinomas (4). Mutations in the RAS genes or PAX8/PPARγ rearrangement occur in ~70% of follicular thyroid carcinomas and with lower frequency in oncocytic (Hürthle cell) carcinomas (5). Regarding the specificity of these mutations for cancer, BRAF V600E mutation and RET/PTC and PAX8/PPARγ rearrangements are overall specific for malignancy in the thyroid, although they have been reported with a very low frequency in benign thyroid lesions (6). RAS mutations occur in malignant and benign thyroid tumors, being found in ~40-50% of follicular and anaplastic carcinomas, 30-40% of follicular adenomas and 10-15% of papillary carcinomas (6).

- Adeniran AJ, et al. Correlation between genetic alterations and microscopic features, clinical manifestations, and prognostic characteristics of thyroid papillary carcinomas. Am J Surg Pathol 2008; 30:216-222.
- Xing M, et al. BRAF mutation predicts a poorer clinical prognosis for papillary thyroid cancer. J Clin Endocrinol Metab 2005; 90:8373-8379.
- Elisei R, et al. BRAF V600E mutation and outcome of patients with papillary thyroid carcinoma: a 15-year median follow-up study. J Clin Endocrinol Metab 2008; 93:3943-3948.
- Lee X, Gao M, Ji Y, Yu Y, Feng Y, Li Y, Zhang Y, Cheng W, Zhao W. Analysis of differential BRAF(V600E) mutational status in high aggressive papillary thyroid microcarcinoma. Ann Surg Oncol. 2009 Feb;16(2):240-5.
- Nikiforova MN, et al. RAS point mutations and PAX8-PPARγ rearrangement in thyroid tumors: Evidence for distinct molecular pathways in thyroid follicular carcinoma. J Clin Endocrinol Metab 2003; 88: 2318-28.
- Nikiforov YE. Recent developments in the molecular biology of the thyroid. In: Lloyd RV, ed. Endocrine Pathology: Differential Diagnosis and Molecular Advances. Humana Press, Totowa, 2004. 191-209.

#### MUTATIONAL ANALYSIS:

For paraffin-embedded surgical specimens, manual microdissection was performed to collect tumor tissue. Specimens with the minimum of 50% of tumor cells in a microdissection target are accepted for the analysis. Optical density readings were obtained. Real-time PCR was performed on the LightCycler platform to amplify BRAF codons 598-601, NRAS codon 61, HRAS codon 61, and KRAS codons 12/13 sequences. Post-PCR melting curve analysis was used to detect possible mutations. Single-step real-time RT-PCR was performed to amplify the fusion points of RET/PTC1 and RET/PTC3 rearrangements or, if required, the mutation type was confirmed by Sanger sequencing of the PCR products.

Amplification of the KRT7 genes was used to control quality and quantity of RNA and amount of epithelial cells present in the specimen. DNA or RNA from samples positive for each of these mutations were used as positive controls. Amplification at 35 cycles or earlier was considered sufficient for this analysis. The limit of detection is approximately 10-20% of alleles with mutation present in the background of normal DNA and RNA.

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| Criteria                       | Yes       | No                      |
|--------------------------------|-----------|-------------------------|
| Diagnosis Discrepancy          |           |                         |
| Primary Tumor Site Discrepancy |           |                         |
| HIPAA Discrepancy              |           |                         |
| Prior Malignancy History       |           |                         |
| Dual/Synchronous Primary Nodes |           |                         |
| Case is (circle):              | QUALIFIED | DISQUALIFIED            |
| Reviewer Initials              | BLH       | Date Reviewed 9/21/2012 |